Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/23/2006
PI Name:	Czeisler, Charles A. M.D., Ph.D.		
Project Title:	Circadian Entrainment, Sleep-Wake Regulation and Performance during Space Flight		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Human Performance Factors, Sleep, and Chronobiology Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	caczeisler@rics.bwh.harvard.edu	**Fax:**	FY 617-732-4015
PI Organization Type:	UNIVERSITY	**Phone:**	617-732-4013
Organization Name:	Brigham and Women's Hospital/Harvard Medical Center		
PI Address 1:	Division of Sleep Medicine		
PI Address 2:	221 Longwood Ave., Ste. 438		
PI Web Page:			
City:	Boston	**State:**	MA
Zip Code:	02115-5804	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	10/01/2000	**End Date:**	02/29/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kronauer, Richard (Harvard University) Gronfier, Claude (Brigham and Women's Hospital / Harvard Medical School) Wright Jr., Kenneth (Brigham and Women's Hospital / Harvard Medical School)		
Grant/Contract No.:	HPF.002.02		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The purpose of the study was to test three specific hypotheses aimed at evaluating entrainment of the human circadian pacemaker to longer-than-24-hour days. These hypotheses were based on the results of our preliminary data which indicate that: (a) the period of the human circadian pacemaker after release from entrainment to the 24-hour day is near to but on average slightly longer than-24-hours, (b) the 24.65-h Martian day length is outside the range of entrainment of the human circadian pacemaker in the presence of a weak environmental synchronizer, and (c) intermittent exposure to bright light is a cost effective means of resetting the human circadian pacemaker with respect to power use and astronaut time compared to continuous exposure to light. We have designed a 65-day long between subjects randomized clinical trial to test the three hypotheses. Twelve subjects were randomly assigned to either Cell A, or B, or C. Subjects assigned to Cell A and Cell B were studied in a longer-than-24-hour sleep-wake schedule in approximately 25 and 100 lux in the angle of gaze during wakefulness, respectively. Subjects assigned to Cell C were studied in a longer-than-24-hour sleep-wake schedule in approximately 25 to 100 lux in the angle of gaze during wakefulness, but in addition, they were exposed to two brief 45 minute pulses of bright light (approximately 10,000 lux) in the evening, separated by 1-h of approximately 100 lux, as a countermeasure to entrain their pacemaker to the longer-than-24-hour day. We anticipated that subjects in cell A and B would fail to appropriately entrain their circadian pacemaker to the longer-than-24-hour day, resulting in sleep and endocrine disruption and impaired alertness and performance. We anticipated that subjects in Cell C, that is, scheduled to receive two relatively brief (45 minutes) daily exposures to evening bright light (approximately 10,000 lux) as a countermeasure to circadian misalignment, would establish a normal entrained circadian phase. Our preliminary results indicate that subjects exposed to the two evening episodes of bright light demonstrate entrainment of their melatonin rhythms to the imposed sleep-wake cycle. They also reveal that lighting levels of approximately 25 lux (comparable to the intensities measured in the middeck for STS-95) were insufficient to maintain a normal phase angle of entrainment, and that approximately 100 lux (comparable to the intensities measured in the Spacehab for STS-95) was also insufficient to maintain a normal phase angle of entrainment in all subjects. Therefore, our preliminary results indicate that two relatively brief (45 minutes each) exposures to bright light in the evening are effective in maintaining entrainment of the circadian pacemaker to longer-than-24 h days, and would be effective as a countermeasure to circadian misalignment during space flight. Our data also suggest that ordinary room light (100-150 lux in the angle of gaze) might be sufficient to maintain entrainment to the 24.65-h solar day of Mars in a small number of astronauts, although at an abnormal phase angle. Both the model and our preliminary data indicate that those astronauts who will have the greatest difficulty adapting to the 24.65 h Martian day will be those with an endogenous circadian pacemaker that has a period shorter than 24 h, which we estimate represents approximately 25 per cent of the population. The results of our research could have a profound effect on the health, productivity and safety of astronauts during an exploration class mission to Mars. On earth, this work has important implications for the treatment of circadian rhythm sleep disorders, such as advanced sleep phase syndrome and shift-work dysomnia, which are anticipated to have a high incidence and prevalence during exploration class space missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Optimal human performance during space flight requires astronauts to maintain synchrony between the circadian pacemaker which regulates the timing of sleep, endocrine function, alertness and performance and the timing of the imposed sleep-wake schedule. Operational demands of space flight necessitate that humans live on day lengths different than the 24-h solar day of Earth. Due to orbital mechanics, astronauts are commonly scheduled to the near equivalent of a shorter-than-24-hour day length in Earth orbit on space shuttle missions; moreover, they will be scheduled to the 24.65-h solar day of Mars on the planned exploration class mission to Mars. Through current support from the NSBRI and NASA, we have demonstrated that a scheduled dim light-dark rest-activity cycle, with a dim ambient light intensity similar to that used aboard the space shuttle middeck, is able to entrain most, but not all human subjects to a scheduled 24-hr day, whereas none of the human subjects scheduled to a 24.6-h day (the period of the axial rotation of Mars) were entrained to this weak synchronizer (see preliminary results). Circadian phase misalignment to the 24.6-h day resulted in sleep disturbance (reduced sleep efficiency), endocrine disturbance (secretion of the sleep-promoting hormone melatonin during the waking day, reduced nocturnal growth hormone secretion and reduced cortisol levels), and impaired daytime alertness and neurobehavioral performance (reduced vigilance). The degree of circadian misalignment to the 24.6-h day was found to be strongly dependent upon the period of each subject-s circadian pacemaker, such that subjects with periods shorter than 24.0 hr demonstrated the greatest degree of circadian misalignment to the 24.6-h day. Due to concerns over high radiation exposure during the voyage to Mars and while on the planet-s surface, NASA engineers have indicated that neither the spacecraft nor the Martian habitat may have windows. Our data suggest that most if not all astronauts would exhibit circadian misalignment if the space flight lighting conditions of approximately 10 lux on the windowless middeck of the space shuttle were present on the space craft while en route to Mars or on the Mars station during their approximately 540 day stay on Mars. The data collected during the past four years of support from the NSBRI demonstrate the need to develop effective and attainable countermeasures to prevent circadian misalignment during an exploration class mission to Mars. Our preliminary results suggest that exposures to bright light episodes of relatively short duration are effective in resetting the human circadian pacemaker and may provide an effective countermeasure to prevent circadian misalignment during exploration class space missions. This work has important implications for the treatment of circadian rhythm sleep disorders, such as advanced sleep phase syndrome and shift-work dyssomnia, which are anticipated to have a high incidence and prevalence during exploration class space missions. Careful analysis of the efficacy of brief, intermittent bright light episodes in the treatment of these conditions has important ramifications for the practical application of bright light treatment during space flight, since repetitive uninterrupted exposure to bright light for many hours each day would not be feasible due to the cost of generating power and other operational demands of space missions. The results of this research could have a profound effect on the health, productivity and safety of astronauts during an exploration class mission to Mars.
Task Progress:	No progress report this reporting period.
Bibliography Type:	Description: (Last Updated: 11/17/2025)